Sumedh Ramakrishna

135, Schwetzinger Str., Heidelberg, Germany
M.Sc. Economics, Heidelberg University (Alfred Weber Institute)
sumedhramakrishnal@gmail.com | LinkedIn | +49 15231629556



Profile

Quantitative Economist specializing in financial risk analytics, econometric modeling, credit decisioning logic, and applied machine learning. Well versed in SQL database extraction, cash-flow and panel data analysis, risk-adjusted portfolio metrics, and statistical underwriting models.

I am **eligible to work** in the following countries without visa sponsorship: Germany, India, United Kingdom (2-year HPI visa), Netherlands (1-year Orientation visa).

Education & Certifications

Heidelberg University *11/2025*
M.Sc. Economics (Grade: 2.0 - Good)

Shiv Nadar University *06/2021*
B.Sc. (Research) in Economics

Research & Analytical Experience

Master's Thesis (Grade: 1.8) *07/2025 – 11/2025*
Alfred Weber Institute (Prof. Stefan Klonner) Heidelberg, Germany

- Analyzed household investment behavior and financial commitment decisions using complex microdata panels from the Indonesian Family Life Survey (IFLS).
- Engineered a quantitative indexing method to structure multi-round longitudinal data, implementing non-linear econometric regressions and causal inference to evaluate household resource allocation under financial constraints.
- Executed reproducible data-cleaning and transformation pipelines in R and Stata across tens of thousands of survey records.

Master's Seminar (Decision Theory & Predictive Modeling) *04/2024 – 08/2024*
Alfred Weber Institute (Dr. Sebastian Ebert) Heidelberg, Germany

- Modeled choice behavior and risk-preference shifts across asymmetric risk environments, evaluating predictive accuracy of contextual choice models against standard expected-utility benchmarks.
- Structured decision trees and non-linear classification models to benchmark risk profiles and identify non-standard decision patterns relevant to credit risk assessment.

Research Intern *05/2021 – 11/2021*
Madras Inst. of Dev. Studies (Dr. L. Venkatachalam) Chennai, India

- Conducted statistical valuation and cost-benefit estimations for regional ecological assets to support structured policy and public investment allocations.
- Synthesized multi-source environmental and survey datasets into structured regression models and executive analytical reports.

Bachelor's Thesis *08/2019 – 05/2020*
Shiv Nadar University (Dr. Arka Roy Chaudhuri) Greater Noida, India

- Investigated macroeconomic liquidity shocks and informal cash circulation anomalies using large-scale National Sample Survey (NSS) expenditure microdata.
- Identified transient consumption spikes using difference-in-differences and time-series controls, evaluating systemic liquidity shifts.

Technical Skills

- **Programming & Databases:** SQL, Python, R, Stata, Git/GitHub, LaTeX, Advanced Excel.
- **Machine Learning & Quantitative Methods:** Supervised/unsupervised learning, regularized regression (LASSO/Ridge), logistic regression, tree-based models (Random Forest, Gradient Boosting), causal inference, panel econometrics, hypothesis testing, portfolio optimization, downside risk metrics (VaR/CVaR), credit risk scoring, factor models.

Languages

- **English:** Native / Bilingual
- **German:** Learning B1
- **Tamil:** Native / Bilingual